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WHEEL AND TIRE ASSEMBLIES – ASSEMBLY PROCESS REQUIREMENTS

TABLE OF CONTENTS

1 GENERAL

1.1 Purpose

This standard describes the minimum requirements for the wheel / tire assembly process and equipment for Chrysler and Fiat production. The wheel / tire module supplier is expected to keep current on new technology that improves efficiency and / or reduces variation in the wheel / tire assembly process. The wheel / tire module supplier is expected to propose new technology for the process as the business case dictates.

1.2 Coverage of this Standard (Applicability) and Limitations on Usage)

This standard applies to all Fiat or Chrysler assembly plants that perform wheel and tire assembly process and/or all other outside companies supplying wheel and tire assemblies to Chrysler or Fiat.

1.2.1 Line Capability: Wheel and Tire Sizes

The wheel / tire assembly line and all pertinent equipment must be capable of handling the entire range of wheels and tires currently released or proposed for the program. The process must be capable of handling (at a minimum) an increase of 5” in rim width and diameter compared to the smallest released rim.

1.2.2 Inventory Control

Throughout the entire wheel / tire assembly process, a FIFO method of inventory control must be used to facilitate clean points for component design or specification change. Assembly lines (internal or external to the vehicle assembly plant) must not process tires that are older than 52 weeks past the DOT code unless they receive written concurrence from Engineering. If a wheel / tire assembler cannot maintain a FIFO system, they must notify Chrysler or Fiat Engineering and Supplier Quality Engineering in writing, noting specifically which components cannot be controlled and the reason why they cannot be controlled. Any exceptions must be approved in writing by Fiat/Chrysler Engineering and Supplier Quality Engineering.

2 REFERENCES

Table 1 - References

Document Number	Document Title
AS-10136	WHEEL AND WHEEL TRIM
CS-9003	SUPPLIER REQUIREMENTS FOR VEHICLE AND SERVICE PARTS: MATERIAL CONTENT REPORTING, MARKING, AND RECYCLABILITY
CS-9800	GENERAL INFORMATION
CS-9801	QUALITY REQUIREMENTS
MS-9990	LUBRICANT - ASSEMBLY AID - WATER BASED
PF-4092	WHEEL BALANCE WEIGHTS - CLIP-ON
PF-13250<S>	STEEL AND ALUMINUM ROAD WHEEL STRENGTH REQUIREMENTS
PF-8983	BONDED COMPONENTS - SURFACE MATCH VERIFICATION METHOD
PF-10510	ADHESIVE BALANCE WEIGHTS

Table 1 - References

Document Number	Document Title
PF-11894<S,D>	TIRES - PRODUCTION TIRE QUALITY
PF-Safety<S>	PRODUCT SAFETY - USE OF SAFETY SHIELDS <S>
19330	WHEEL BALANCE WEIGHTS (TPM)
19331	WHEEL BALANCE WEIGHTS (LEAD FREE) (TPM)
7-G0112	THREADED FASTENER TIGHTENING TEST ON COMPLETE SYSTEMS AND VEHICLES (PGE)
9.01102	QUALITY OF SUPPLIES FIAT GROUP AUTOMOBILES S.P.A. (CFO)
9.01398	TYRES FOR PASSENGER CARS AND DERIVED VEHICLES (CME)
9.01399	RUBBER VALVES FOR TIRES USED ON STEEL AND LIGHT ALLOY RIMS FOR VEHICLES, DERIVATIVES AND COMMERCIAL VEHICLES (CME)
9.02350	STEEL WHEELS FOR CARS, DERIVATIVES AND LIGHT COMMERCIAL VEHICLES (CME)
9.02350/01	LIGHT ALLOY WHEELS FOR CARS, DERIVATIVES AND LIGHT COMMERCIAL VEHICLES (CME)
9.02352	ADHESIVE BALANCE WEIGHTS FOR ALUMINUM WHEEL (CME)
9.02352/01	CLIP ON BALANCE WEIGHTS FOR STEEL WHEELS (CME)
9.96465	INFLATION PRESSURE MONITORING SENSOR (TPMS) WITH VALVE FOR TIRES (CEL)
PS-11343	TIRE PRESSURE MONITOR (TPM) SYSTEM ASSEMBLY (CORAX MODULE AND CAN BUS ARCHITECTURE WITH UDS DIAGNOSTICS)
Quality and Reliability Documents	
Other Documents	
Tire and Rim Association, Inc. Yearbook	
European Tire and Rim Technical Organization	

3 DEFINITIONS/ABBREVIATIONS/ACRONYMS/SYMBOLS

Conicity: Lateral force generated by a tire due to tire manufacturing variation such as off center belt ply etc.
Correction Plane: Inboard or outboard location of the balance weight.
ETRTO: European Tyre and Rim Technical Organization
FIFO: First In, First Out
Imbalance - Couple: A condition where the inertial axis of the tire / wheel assembly is not parallel to, but intersects the axis of rotation. This condition causes steering wheel vibration and wheel wobble.
Imbalance - Dynamic: An imbalance condition consisting of static and couple unbalance. This condition is corrected by placing balance weights on both inner and outer wheel flanges.
Imbalance - Static: A condition where the inertial axis of the tire/wheel assembly is parallel to, but does not lie on the axis of rotation. This condition causes vehicle shake and wheel hop vertically.
Imbalance - Residual: The measure of imbalance that remains in the tire / wheel assembly after balance weight application, expressed in terms of an equivalent mass (oz.) located at that wheel's rim flange (in.).
Match Mounter: Indexes tire force high point to wheel average radial first harmonic low point.

Mounter: Mounts tire onto wheel.
Pilot Bore: Center Bore of the wheel that pilots on the hub.
T & RA: Tire and Rim Association
Tire Bead Soaper: Applies lubricating soap to tire beads.
Tire Radial First Harmonic: Defined as the first harmonic component of the radial composite force variation.
Tire/Wheel Inflator: Inflates wheel/tire assembly.
TPM: Tire Pressure Monitor
Unidirectional tire: Defined by the tread design which is designed to rotate in one specific direction as indicated on the sidewall.
Valve Stemmer: Installs the tire valve into the wheel.
Wet-out: The physical contact between an adhesive surface and the substrate.
Wheel Beadseat Soaper: Applies lubricating soap to wheel beadseat(s).
Wheel Average Radial First Harmonic (AR1H): The vector average of the inboard bead and outboard bead radial 1st harmonics.

4 PROCESS AND PROCESS CONTROL REQUIREMENTS

4.1 Condition of the part prior to processing

Fiat/Chrysler is responsible for conformity of components delivered on consignment by Fiat/Chrysler.

For components not supplied by Fiat/Chrysler, the wheel/tire assembly supplier is responsible for procurement and may pick the manufacturers of components following approval and qualification by Fiat/Chrysler.
For Fiat approved balance weights refer to specs 19330 and 19331.
Rubber snap-in valves should be compliant to ETRTO standard and spec 9.01399.
Component manufacturers are responsible for compliance with component specifications. The supplier of the complete wheel/tire assembly is responsible for the assembly process and therefore for defects due to the process itself and for the correct use (handling/storage) of the components to be assembled.

4.1.1 Inbound Tires

Inbound tires are to be loaded onto the assembly line in matching conicity sets (or matching axle sets where applicable) with the match mount sticker or ink mark on the outboard side. Tires will be marked with a combination of stickers and/or paint dots on the external/outboard sidewall of the tire. Tires may also be marked with semi-permanent stripes across the tire tread. It is the responsibility of the wheel / tire assembler to understand each tire manufacturer's methods for marking conicity and manage tire stock accordingly. The module supplier's process must allow tires to stabilize at room temperature before being mounted on a wheel. Tires must be stored in clean conditions free from exposure to sunlight or strong artificial light, heat, ozone (electrical machines) and hydrocarbons per ETRTO Recommendations on Storage section S8, S9 and S10.

NOTE: Unidirectional tires must be assembled and shipped in a manner that supports the vehicle assembly plant's wheel / tire assembly installation process. In a given vehicle set, the wheel / tire assembler must be capable of identifying and controlling the proper orientation of each tire (clockwise [passenger side] or counterclockwise [driver side]). Fiat/Chrysler Engineering to provide direction on the pattern in which wheel / tire assemblies are installed onto vehicle within the vehicle assembly plant. (For example, a specific Assembly Plant may install the four sequential assemblies in a given vehicle set in the following order: left front, right front, left rear, right rear).

4.1.2 Inbound Wheels

Inbound wheels are to be loaded onto the line ensuring that the wheels do not hit each other while on the line prior to tire mounting, to prevent wheel damage.

4.2 Vision System Verification

Wheel / Tire Assembler must ensure that the Vision System has the ability to automatically distinguish between the different tires and wheels, including the ability to distinguish between different wheel finishes on the same wheel and tires that are the same size and brand, but have different constructions (will require the tire supplier to use semi-permanent identification markings that are compatible with the Vision System).

4.3 Tire Pressure Monitor (TPM) / Valve Stem Installation

Wheel / Tire assembler must be capable of installing TPMs, rubber valve stems and metal valve stems (for heavy duty Truck program) into both steel and aluminum wheels using Chrysler/Fiat approved tooling and procedures. In the event that non-lubricated tire valves appear in the assembly plant, use the tire lubricant specified in Section 4.5.1 prior to installation of the tire valve in the rim (valve stems must be re-lubricated if not installed into a wheel within 1 hour of applying initial lubrication).

4.3.1 TPM with mounting Nut

Refer to TPM spec PS-11343 or 9.96465. A residual torque confirmation is required. The frequency will be defined in the process control plan.
For tightening torque verification method, refer to spec 7.G0112.

4.3.2 Snap-in TPM

Refer to TPM spec PS-11343.

4.3.3 Metal Valve Stem

From the releases, identify the correct metal valve stem for the wheel being built. Remove the valve stem cap (if required). Remove the purchased in assembly nut (if required). Check for presence of o-ring or grommet. Insert metal valve stem through the wheel valve hole so that the valve stem cap end is visible from the outboard side of the wheel. Obtain and hands start the valve stem nut. Torque the nut to the value provided on the released wheel/tire assembly drawing, AMPS sheet or Fiat vehicle specific torque document. Re-install the cap or install the valve stem extension as required. Installation operation must not damage valve stem or extension. During nut tightening, the valve shall not rotate. In case of rotation, dismount the metal valve stem and replace it with a new one (including o-ring or grommet).
For tightening torque verification method, refer to spec 7.G0112.

4.3.4 Rubber Valve Stem

Using Chrysler/Fiat approved valve stem installation tooling, the operator must insert rubber valve stem through the wheel valve hole so that the valve stem cap is visible from the outboard side of the wheel. Installation operation must not cut or damage valve stem.

4.4 Wheel Orientation

Prior to entering the tire mounting station, wheels may need to be oriented specific to TPM requirements so as not to damage the TPM during tire mounting and inflation snap-up. Chrysler/Fiat engineering will provide direction based on input from TPM supplier. Refer to TPM spec PS-11343.

4.5 Soaping

Assembler must soap the tires in such a manner that the entire tire bead (from heel to toe) is uniformly soaped, 360 degrees around each upper and lower bead. Refer to Figure 1.

Assembler must soap the wheel's inboard and outboard bead areas, from inboard of the safety humps and up the flange walls. Refer to Figure 1.

It is the assembler's responsibility to select a system that meets the requirements for each size wheel / tire combination in the program's lineup.

Assembler must be able to directly monitor and adjust the quantity of soap used for each wheel and tire component. An alarm or warning system is required to alert the required personnel that the soap was not applied directly to the component.

NOTE: During any downtime, all wheels and tires that have been soaped but sit prior to the inflation station MUST be purged from the soaping station if the line has stopped for more than 5 minutes. Soaped wheels / tires must either be processed through the inflation station, or removed from the line. Any exceptions must be approved in writing by Chrysler/Fiat Engineering and SQA.

4.5.1 Approved Soap

Wheel / Tire assembler must use undiluted P3 Tire Lube 40 (ref. Henkel p/n 643711), undiluted Clean EP 60 or Mecafluid S263P diluted in 10% soap and 90% water.

4.6 Tire Mounting

Single bead mounting is required. Any deviations must be approved in writing by Chrysler/Fiat Engineering. Mount the tire on the rim, being careful not to scuff the tire liner at the bead toe. Always mount the tire on the wheel from the rim side that is closest to the drop center at its deepest point. (See Figure 1)

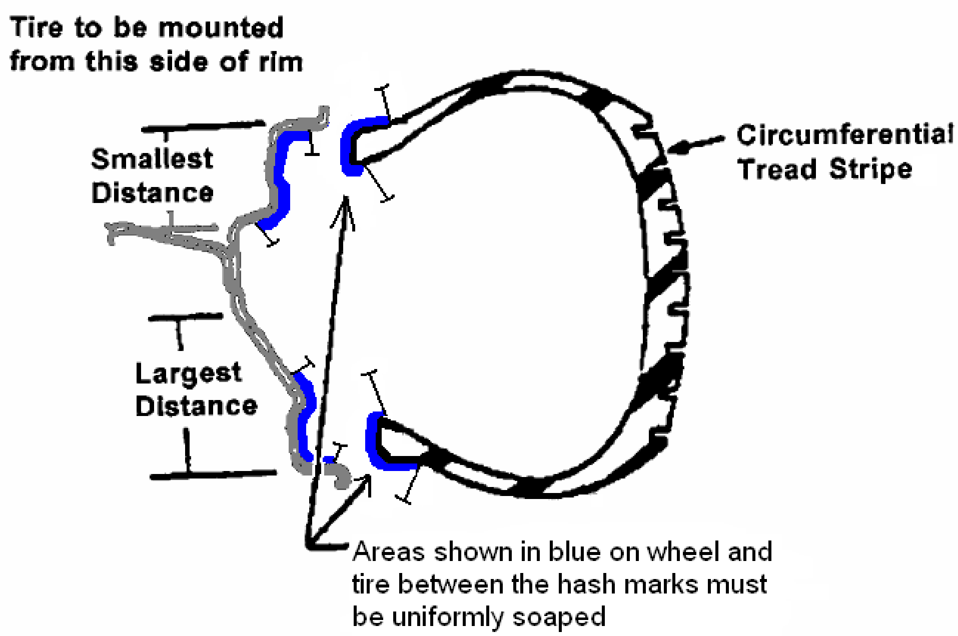

Figure 1 - Wheel And Tire Soap Coverage / Mounting Orientation

4.7 Match Mounting (if applicable)

The match mounting machine must scan both tire and wheel; identify the tire uniformity high point or imbalance low point indicator (typically a paint dot or sticker) and wheel uniformity low point or imbalance high indicator (typically a sticker), and orient the tire and wheel marks to within 25mm of each other's geometric center when measured along the wheel rim flange unless otherwise directed by Chrysler/Fiat Engineering (Refer to Figure 2). Match mount equipment must be capable of reading the wheel and tire marks.

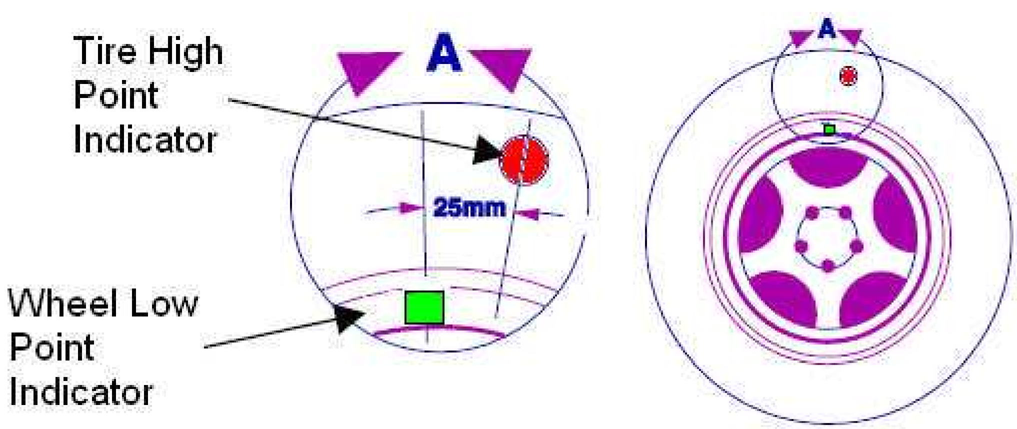

Figure 2 - Wheel/Tire Match Mounting Orientation

The match mounter must be able to identify purposely unmarked wheels or tires and allow those to pass through.

4.7.1 Marking Indicator Specifications

Requirements for the low point marking indicators of wheels and the high point marking indicators of tires are specified in PF-13250<S> , 9.02350 , 9.02350/01 (Wheels) and PF-11894<S,D> , 9.01398 (Tires).

4.7.2 Missing / Replacement Stickers (if applicable)

All tires or wheels that are missing high or low point stickers at any stage in the wheel / tire assembler's process must be removed from the line and quarantined until a tire or wheel supplier representative is notified and can replace the stickers themselves. The wheel / tire assembler is not authorized to replace the stickers unless properly trained by the appropriate tire or wheel supplier and this re-marking process must be approved by Chrysler/Fiat Engineering and SQA (Evidence of training and visual aids must be presented to Chrysler/Fiat as part of the approval process). Match mount stickers must be present on all tires and wheels when leaving the wheel / tire assembler as these stickers are required to provide each plant a visual audit for conicity and to ensure matching conicity sets for each vehicle.

4.8 Inflation

Air free from products which could damage the assembly (e.g: oil, metallic particles, etc.) must be used for inflating the tires.

4.8.1 Inflation Pressure

Tire inflation pressures are specified by Fiat / Chrysler Engineering in the released assembly drawing or AMPS and are considered as cold tire inflation pressures. If tire inflation pressures are not defined by Chrysler/Fiat Engineering, Table 2 below is to be used as a default.

Table 2 - Tire Inflation Pressures

Tire Type	Pressure Kpa	Pressure PSI
P-metric	240	35
P-metric Extra Load	280	41
European Metric (Standard Load & Extra Load)	300	43.5
LT – Load Ranges B,C	280	40
High Pressure Spare – S Type	280	41
High Pressure Spare – T Type	414	60
LT – Load Ranges D, E	428	62
Light Duty Commercial Vehicle (P Segment)	400	59
Light Duty Commercial Vehicle (LCV Segment)	450	65
Light Duty Commercial Vehicle (Camper)	550	80
MDT (Medium Duty Truck) – Load Ranges F, G	660	95

Tire inflation pressures must be maintained within - 0 kPa / + 42 kPa, (- 0.0 psi / + 6.0 psi) for assemblies with inflation pressure up to 400 kPa (59 psi) and within - 0 kPa / + 70 kPa (- 0 psi / + 10 psi) for assemblies with inflation pressure over 400 kPa (59 psi). The inflation system must be capable (per Chrysler/Fiat PSO manual) of inflating tires from 190 kPa - 690 kPa. (28 psi - 100 psi). Tires may need to be inflated to the maximum pressure indicated on the tire sidewall. Direction will be provided by Fiat / Chrysler Engineering.

4.8.2 Maximum Pressure Differential for Each Vehicle Set of Tire and Wheels

For every sequenced vehicle set of tire and wheels, a maximum pressure differential of 30 kPa, (4.0 psi) must be maintained. Spare tire and wheel assemblies are excluded. For vehicles with dual pressures, then pressure differential applies only per axle pair.

4.8.3 Tire Inflation Pressure for Medium Duty Truck Tires (All tires with Steel Carcass Plies)

This type of tire utilizes steel cords in the sidewall. As such, they cannot be treated like normal light truck tires. Adjusting tire pressure must be performed by personnel trained, supervised and equipped accordingly.

4.8.4 Inflation Pressure Adjustment for Medium Duty Truck Tires (All tires with Steel Carcass Plies)

When adjusting the inflation pressure of a tire and wheel assembly, use the Remote Mount Automatic Inflation Kit and a Tire Inflation Cage similar to model number #174-2240 available from TEAM PSE (PENTASTAR Service Equipment - 1-800-223-5623). All persons must be clear of the trajectory area.

4.9 Valve Stem / TPM and Tire to Wheel Air Seal Integrity Confirmation

Wheel / Tire Assembler must ensure the integrity of the Valve Stem / TPM and tire to wheel air seal. When introducing a new Valve Stem / TPM, Tire or Wheel, it is the responsibility of the wheel / tire assembler to spray a soap and water solution on the Valve Stem / TPM to Wheel interface and the Tire to Wheel interface and visually inspect for air bubbles. This inspection is to continue for 1 week beyond introduction of new component(s).

4.10 Bead Seater / Load Simulator

A bead seater or load simulator is mandatory. All wheel and tire assemblies must be processed through the bead seater / load simulator equipment unless directed by Chrysler/Fiat Engineering (typical exception may be for compact spare assemblies). The bead seater / load simulator equipment must engage both the inboard and outboard side of the tire and the wheel assembly and must complete a minimum of one full revolution for each wheel and tire assembly. It is the responsibility of the wheel / tire assembler to ensure that the tire beads are fully seated on the wheel. Wheel / tire assembler must actively participate with the supplier of bead seater / load simulator equipment and tire and wheel suppliers to ensure proper bead seating. If the bead seat / load simulator equipment becomes inoperable, the module supplier must notify Chrysler/Fiat Engineering and SQA in writing, noting specifically which modules will not be processed through the equipment and the reason why the equipment is down.

4.11 Uniformity (Road Force Variation)

Specifications for uniformity (assembly force variation) are defined and released by the Fiat / Chrysler Release Engineer on the module assembly drawing. All wheel / tire assemblies shipped into the vehicle assembly plant must conform to the released uniformity specifications. It is the responsibility of the wheel / tire assembler to work with Fiat / Chrysler Engineering in order to help develop appropriate screening specifications.

4.11.1 Uniformity (Road Force Variation) Measurement Conditions:

Loading conditions:
Loading conditions must be equivalent to what is shown on Chrysler/Fiat released assembly drawing or AMPS and may be unique for each tire.

Inflation Pressure:
See section 4.8 for tire and wheel assembly.

4.11.2 Uniformity (Road Force Variation) Equipment Accuracy:

The wheel / tire assembly force variation equipment must be capable of automatically locating and marking the radial first order harmonic to the following levels of accuracy shown below in Table 3:

Table 3 - Uniformity Marking Accuracy

Assembly 1st Order Harmonic	Maximum Error (Degrees)
0 to 22 N (0 to 5 lbs.)	+/- 180
>22 to 45 N (5 to 10 lbs.)	+/- 45
>45 to 67 N (>10 to 15 lbs.)	+/- 30
>67 N (>15 lbs.)	+/- 25

The equipment must be capable of a mean marking error which is less than half the maximum error.

The equipment must be capable of marking the location of the assembly's radial first harmonic high point location on the sidewall of the tire, and distinguish between in specification or out of specification assemblies. The wheel / tire assembler must be able to provide assembly uniformity data on all out of specification assemblies. Assembly uniformity data must be printed on a label and attached to both the wheel and tire of all out of specification assemblies. Complete data must be recorded for each rejected component (itemized for each pass through the process) and available for review by Fiat / Chrysler Engineering, wheel suppliers, and tire suppliers. Deviation from this identification process of tire and wheel must be approved by Fiat / Chrysler Supplier Quality and Engineering.

4.12 Conicity Measurement Capability

The uniformity equipment must be capable of measuring assembly conicity magnitude and direction.

4.13 Insertion of New Assemblies

If an assembly is reinserted to the line at any point in the assembly process, the wheel / tire assembler must be able to provide traceability data by VIN for all replacement assemblies. The replacement assembly must be within specification.

4.14 Balance

All wheel / tire assemblies must be processed through primary and audit balancers unless directed by Fiat / Chrysler Engineering (typical exception may be for compact spare assemblies) and must conform to released residual imbalance specifications shown on the module assembly drawing. Assemblies that require correction at the audit station must be re-measured after the audit weight is applied to verify conformance to released residual imbalance specifications.

4.14.1 Primary / Audit Balancer

The primary and audit balancers must:
- Measure dynamic imbalance of the tire / wheel assembly (per correction plane) by balancing the wheel / tire assembly centered about its pilot bore axis.
- Calculate an optimized static and couple imbalance (in agreement with Table 4) and determine amount and location of balance weights to be installed on flanges (inboard, outboard) or mid-plane

locations. Calculations must use balance weight locations called out on the released module assembly drawing.
- Use a minimum of an 8-chuck expanding collet and must be a multiple of the number of bolt holes.
- Use a double chucking operation (chuck + unchuck) prior to balancing to settle the assembly on the collet's flange prior to balancing.
- Mark weight locations on tire sidewall with a semi-permanent automatic marking system (unless a manual operation is allowed by Chrysler/Fiat Engineering).
- In accordance with Table 4, reject all uncorrectable assemblies (unless it is feasible to apply the maximum allowable weight at the primary balancer in order to attempt to correct the assembly at the audit balancer using allowable audit maximum weight).
- Assembly balance data must be printed on a label and attached to both the wheel and tire of all out of specification assemblies. Complete data must be recorded for each rejected component (itemized for each pass through the process) and available for review by Fiat / Chrysler Engineering, wheel suppliers, and tire suppliers Deviation from this identification process of tire and wheel must be approved by Fiat / Chrysler Supplier Quality and Engineering.

4.14.2 Offline Dynamic Balancer

Offline equipment must be able to calculate static and couple (or dynamic) imbalance and support the current and proposed product line. Off-line balancers capable of also measuring wheel/tire assembly uniformity are preferred. This is required for immediate root cause analysis when production equipment is in use.

4.14.3 Wheel to Primary / Audit Balancer Collet Interface Verification

The Wheel/Tire Assembler must ensure the integrity of the interface between the wheel and the primary and audit balancer collets. When introducing a new wheel, it is the responsibility of the wheel/tire assembler to install the production intent wheel with a section removed onto the collet of the primary balancer(s) and audit balancer(s). The Wheel/Tire Assembler will inspect the wheel to collet interface to verify there are no interferences and that the wheel mounts properly to collet(s).

4.15 Residual Imbalance Specifications

Chrysler/Fiat Engineering will provide residual static and residual couple imbalance specifications using the module assembly drawing or AMPS. If no direction is provided default to Table 4 below:

Table 4 - Default Allowable Residual Imbalance By Platform And Production Plant

	NAFTA plants Fiat, Chrysler Passenger Car, Jeep, Minivan, Light Duty Truck	**NAFTA plants** Medium Duty Truck	**EMEA plants** Fiat / Lancia / Alfa Romeo / Jeep Passenger Car, Light Commercial Vehicle P-segment	**EMEA plants** Fiat Light Commercial Vehicle LCV segment	**Maserati**
Minimum Correction Weight	0.25 oz. (7 g)	1.00 oz. (28 g)	5 g (0.18 oz.)	5 g (0.18 oz.)	5 g (0.18 oz.)
Correction Weight Increment	0.125 oz. (3.5 g) up to 3 oz. (85 g)	0.50 oz. (14 g)	Clip-on 5 g (0.18 oz.)	Clip-on 5 g (0.18 oz.) up to 50 g (1.76 oz) then 10 g (0.35 oz.)	Clip-on Not applicable

Table 4 - Default Allowable Residual Imbalance By Platform And Production Plant

	NAFTA plants **Fiat, Chrysler Passenger Car, Jeep, Minivan, Light Duty Truck**	**NAFTA plants** **Medium Duty Truck**	**EMEA plants** **Fiat / Lancia / Alfa Romeo / Jeep Passenger Car, Light Commercial Vehicle P-segment**	**EMEA plants** **Fiat Light Commercial Vehicle LCV segment**	**Maserati**
	then 0.25 oz. (7 g)			up to 100 g (3.53 oz.)	
			Adhesive 5 g (0.18 oz.)	Adhesive 5 g (0.18 oz.)	Adhesive 5 g (0.18 oz.)
Maximum Correction Weight Per Plane @ Primary	5.00 oz. (142 g)	6.00 oz. (170 g)	Clip on 50 g (1.76 oz.)	Clip on 100 g (3.53 oz.)	Clip-on Not applicable
			Adhesive 80 g (2.82 oz.)	Adhesive 100 g (3.53 oz.)	Adhesive Inner side 18" – 19" 40 g (1.41 oz.) 20" – 21" 60 g (2.12 oz.)
					Adhesive Outer side 18" – 19" 70 g (2.47 oz.) 20" – 21" 90 g (3.17 oz.)
Maximum Correction Weight Per Plane @ Audit	1.00 oz. (28 g)	1.00 oz. (28 g)	None	None	None
Maximum Residual Static Imbalance	0.25 oz. (7 g)	0.75 oz. (21 g)	Not Applicable	Not Applicable	Not Applicable
Maximum Residual Couple Imbalance	0.50 oz. (14 g)	1.00 oz. (28 g)	10 g per plane (0.35 oz.)	15 g per plane (0.53 oz.)	5 g per plane (0.18 oz.)
Maximum Total Per Plane (after Audit)	6.00 oz. (170 g)	7.00 oz. (198 g)	Clip on 50 g (1.76 oz.)	Clip on 100 g (1.76 oz.)	Clip-on Not applicable
			Adhesive 80 g (2.82 oz.)	Adhesive 100 g (3.53 oz.)	Adhesive Inner side 18" – 19" 40 g (1.41 oz.) 20" – 21" 60 g (2.12 oz.)
					Adhesive Outer side 18" – 19" 70 g (2.47 oz.) 20" – 21" 90 g (3.17 oz.)

NOTE 1: A maximum of 2 weights per plane are allowed after the audit balancer except for the outboard plane of wheels that have an exposed clip on weight. Only one weight is permitted on this type of wheel.

NOTE 2: The above table is based on the assumption that balance weights are applied at an MC-flange location for each wheel size. If adhesive weights are to be used, an adjusted imbalance specification should be calculated and implemented by the Fiat / Chrysler Release Engineer

4.16 Wheel Weights and Wheel Weight Installation

Wheel weights must be the correct type for each wheel flange as specified by Fiat / Chrysler Engineering releases and summarized in the released module assembly drawing or AMPS.

4.16.1 Wheel Surface Preparation for Adhesive Wheel Weights

Wipe clean the area of the wheel to which the component is to be bonded with a clean lint-free cloth moistened with a 50/50 (by volume) mixture of isopropyl alcohol and clean water. Wipe the bonding surface dry immediately with a dry lint free cloth before applying the weight. A microfiber lint-free white polyester cloth (Chrysler non-production material part number 35-260-0136) is also acceptable. The surface to which the adhesive weight is to be bonded must be free of dirt, oil, dust, residues, and other foreign material. Be sure not to touch the cleaned surface.

4.16.2 Adhesive Wheel Weight Application Conditions / Requirements

The maximum time between cleaning of the bonding surface and component application to that surface must not exceed 5 minutes. Surrounding local conditions to which the prepared bonding surfaces are exposed must be free of lubricants, releasing agents, over sprays, excessive airborne dirt or dust particles, or any materials or processes which could present contaminants to the bond. At the time of application, the temperature of the adhesive wheel weight and the surface to which it is to be applied shall be a minimum of 10 °C (50 ° F), for adhesive tack, and a maximum of 35 ° C (95 ° F).

4.16.3 Adhesive Wheel Weight Installation

In three minutes or less prior to application of the adhesive wheel weight, peel off the protective backing without touching the adhesive portion. The center of the wheel weight must be applied within +/-12 mm (0.5 in.) of the center of the semi-permanent weight location mark in the radial direction. Cross-car location for the outboard (mid-plane) and inboard is specified on the assembly drawing or AMPS. Cross-car tolerance is +5mm/-0mm. Adhesive weight should be located as far outboard as possible for the outboard flange and inboard as possible for the inboard flange.

When not utilizing a semi-automated application tool, manually apply consistent and uniform force over the entire wheel weight. Be sure to press the adhesive wheel weight firmly, contouring the weight to the wheel.

NOTE: The percent total intimately contacted (wet-out) bond surface area must be at least 80%. It must also exceed the requirements of PF-8983.

4.16.4 Adhesive Balance Weight Wet Out Tool to Wheel Clearance Verification

Wheel/Tire Assembler must verify that there is adequate clearance between their adhesive balance weight wet out tool and the wheel. When introducing a new wheel that requires adhesive balance weights, it is the responsibility of the wheel/tire assembler to inspect for interferences between the wheel and the balance weight wet out tool. No interferences are allowed.

4.16.5 Clip-on Weight Installation

The center of the wheel weight must be applied within +/-12 mm (0.5 in.) of the center of the semi-permanent weight location mark in the radial direction. Fiat / Chrysler recommends using a soft faced hammer (1 ¼" - P/N: 4A111 suggested or equivalent, Supplier: Materials Management 1-800- 545-1794) to install clip-on weights. The instrument (hammer/mallet) and technique used to install the balance weight must not deform the balance weight clip (which will reduce the retention of the balance weight). After installing the weight, the maximum allowable gap between the clip and the flange is 0.6 mm.
Re-use of balance weights is not permitted.

4.16.6 Balance Weight Pull Off / Push Off Test

Assembler is responsible to test the clip-on and adhesive weight application process according to the respective standards PF-4092 or 9.02352/01 (Clip-on) and PF-10510 or 9.02352 (Adhesive).

4.17 Wheel Film Installation (If Required - See Module Assembly Drawing)

Prior to application, the wheel surface must be clean and dry. Use a micro-fiber cloth and 70% Isopropyl alcohol/30% water solution to clean tire lubrication from the A-surface of the wheel only where the wheel film is being applied. There should be a minimum of 80% adhesion of the wheel film to the wheel spoke surface and 100% at the rim flange edge. There should be no excessive wrinkles or blisters due to trapped air. The wheel film placement must maintain a 6mm maximum concentricity to the wheel to provide clearance to the wheel secure tooling in the center and not to leave gaps at the outer edge of the wheel.

4.18 Center Cap Installation (If Required - See Module Assembly Drawing)

Wheel/Tire Assembler must verify that the installation process of the center cap is not causing damage to the wheel or center cap.

4.19 Warning labels for spare wheel assemblies (If Required - See Module Assembly Drawing)

Wheel/Tire Assembler must apply warning labels for spare wheel assemblies if specified by Fiat / Chrysler Engineering. The position and orientation of labels onto the wheel will be identified by Fiat / Chrysler Engineering.
Prior to application, the wheel surface must be clean and dry. There should be no wrinkles or blisters due to trapped air.

5 SPECIAL REQUIREMENTS

5.1 Manual Tire and Wheel Mount and Inflation

Mount the tire on the wheel following the tire changer manufacturer's instructions, paying special attention to the following to avoid damaging TPM sensor.

- Rotating Wheel/Tire Changers – Once the wheel is mounted to the changer, position the sensor valve stem approximately 210° from the head of the changer in a clockwise direction before rotating the wheel (also in a clockwise direction) to mount the tire. Use this procedure on both the upper and lower tire beads.

- Rotating Tool Tire Changers – Position the wheel on the changer so that TPM Sensor is located approximately 210° clockwise from the installation end of the mounting/dismounting tool once the tool is mounted for tire installation. Make sure the TPM Sensor is clear of the lower bead breaker area to avoid damaging the sensor when the breaker rises. Rotate the tool in a counterclockwise direction to mount the tire. Use this procedure on both the upper and lower tire beads.
- Inflate tire and wheel assembly to inflation pressure specified in section 4.8
- Reinstall the correct valve cap
- Reinsert wheel/tire assembly back into the assembly line prior to the bead seater. If the manual mount and inflation process is used, the module supplier must notify Fiat / Chrysler Engineering and SQA in writing, noting specifically which modules will be manually mounted and the reason why they cannot be processed on the automated equipment.

5.2 Out of Specification Assemblies

Any assembly that does not conform to the released specifications must be removed from the wheel / tire assembly line and reworked as outlined below. For all wheel / tire assemblies that cannot be balanced or those that do not pass uniformity screening limits, the wheel / tire assembler must have the capability of replacing the out-of-specification wheel / tire assembly with one within specifications. The replacement assembly must match all technical attributes for the VIN being built including conicity and direction in the case of directional tires.

All out of specification wheel and tire assemblies should be reworked using the following procedure.

5.2.1 Remove Wheel Balance Weights (If Applicable)

To remove adhesive wheel weights use a tool similar to what is shown in Figure 3 below. Perpendicular to the length of the weight, slide the tool under the adhesive portion lifting off two segments at a time while being careful not to damage the wheel. Remove any residual adhesive with mineral spirits or naphtha.
Do not use methyl ethyl ketone (MEK) or acetone as it will damage the clear coat. Do not reuse removed balance weights. Any deviation must be approved in writing by Fiat / Chrysler Engineering.

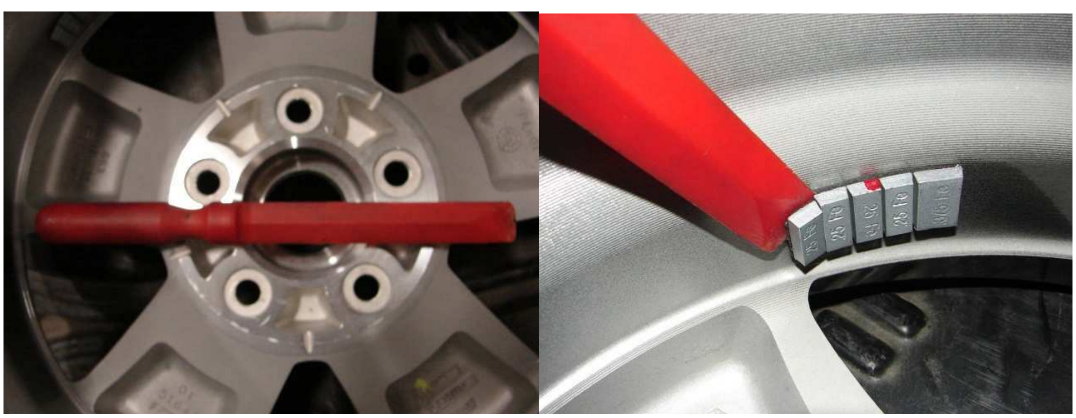

Figure 3 - Adhesive Weight Removal Tool

5.2.2 Deflate Tire

A. Remove valve cap.
B. Depress tire valve core with a deflation tool similar to the one shown below in Figure 4. Tire will then deflate.

C. Reinstall valve cap once tire is deflated

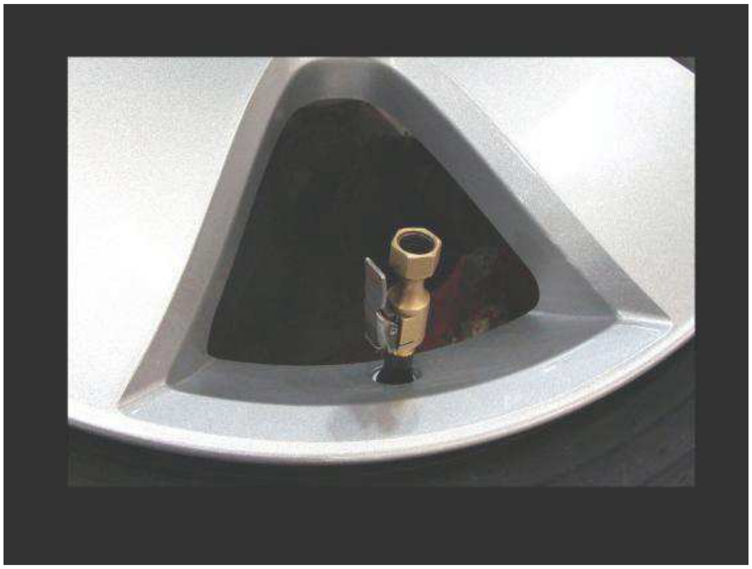

Figure 4 - Tire Deflation Tool

5.2.3 Disassemble

Remove tire from wheel using offline mount and balance equipment using the following procedure:

A. Dismount tire from wheel following tire changer manufacturer's instructions while paying special attention to the following to avoid damaging the pressure sensor.
B. When breaking the tire bead loose from the wheel rim, do not use the Bead Breaker in the area of the TPM Sensor. That includes both upper and lower beads of the tire.
C. After breaking the bead, use tire specified soap (section 4.5.1) to lubricate the assembly prior to dismount to prevent any potential damage.
D. When preparing to dismount the tire from the wheel, carefully insert the mounting / dismounting tool at the valve stem +/- 10° then proceed to dismount the tire from the wheel. Use this process on both the upper and lower tire beads.

Every time a tire has been dismounted from a wheel, the tire and wheel must be inspected per section 7.1.6 for damage.

5.2.4 TPM Sensor Disassembly Procedure

Refer to TPM spec PS-11343.

5.3 Reprocess

Reintroduce components from out of specification assemblies at the beginning of the assembly process. Ensure that reprocessed tires are processed with a different wheel and reprocessed wheels are processed with a different tire.

Wheel / tire assembler must have the capability to direct reprocessed tires to a specific wheel part number. For example, they should be able to direct the reprocess tires to a steel wheel or to an aluminum wheel with an offset flange. Target wheels for reprocess tires will be identified on the released module assembly drawing.

When re-inserting tires from out of specification assemblies back onto the tire line, only 1 previously processed tire may be added to a potential vehicle set unless Chrysler/Fiat Engineering is contacted and provides their concurrence.

Wheel / tire assembler must use a FIFO inventory control method when inserting replacement assemblies and previously processed tires. Tires may be reprocessed a maximum of 5 times.

5.3.1 Alternate Reprocess Procedure

For assemblies that are out of specification for uniformity or unbalance, rotate tire 90 degrees to 180 degrees on wheel, re-soap and inflate to proper inflation pressure per section 4.8. Wheel/tire assembly must be reinserted back into the assembly line prior to the bead seater. The wheel / tire assembly may be re-inflated by the standard assembly process or a manual inflation process. If this option is used, Fiat / Chrysler Supplier Quality and Release Engineering must be notified to investigate why components need to be rotated.

6 QUALITY

Refer to CS-9801 or Fiat 9.01102 for general quality requirements.

The wheel / tire assembler must have a quality assurance program in effect assuring compliance to Fiat / Chrysler's quality requirements specified within the quote package, Fiat / Chrysler's SQA manual, Guide to Quality Systems Requirements, applicable Engineering Standards, and any part drawings. All wheel / tire assemblies must be received at the vehicle assembly plant without degradation of the following parameters:

- Inflation Pressure
- Residual Imbalance
- Radial and/or Lateral Road Force Variation or Run out.
- Overall Appearance (conicity stickers must be on tires).

Neither the tire nor the wheel outside surface shall have any markings other than those on the separate parts as delivered to the assembler or those indicating the upper and lower balance weight locations. Any deviations must be approved by Chrysler/Fiat Engineering, SQA, and the appropriate PVE.

6.1 Inspection Records

Production inspection records (audit results) shall include:

- Date and time or VIN/Sequence number,
- Wheel and tire combination,
- Inspector, and
- Inspection status.

These production inspection records must be maintained on file for the life of the program and available for review at any time. Product Action Notices are to be retained for not less than one calendar year and not more than one calendar year plus 30 days. If the inspection of the safety shielded processes as identified in this standard results in a reject a PAN must be initiated. For all other processes identified in this standard, the appropriate manager must be notified immediately. The appropriate PVE must also be notified immediately for vehicle containment and subsequent corrective action.

6.2 Audit Balancer and Uniformity Machine Records

Audit balancer and uniformity machine data, listed below, shall be recorded at shift end by designated manufacturing personnel. A weekly summary of the data shall be prepared and forwarded to the Fiat / Chrysler Release Engineer and SQA.

- Total assemblies.
- Total good parts.
- Total correctable parts.
- Total uncorrectable parts.
- Cumulative summary of wheels and tires that have been processed the maximum number of recommended times as mentioned in section 4.2.

6.3 Process Conformance

Process conformance shall be tracked daily using Cpk or First Time Capability (FTC). The following equations should be used:

Cpk = (Upper Spec. Limit - Sample Mean Value) / (3 x Sample Standard Deviation)

%FTC Balance = 100 x [Total Parts Processed through Audit Balancer – Uncorrectables] / Total Parts Processed through Audit Balancer

%FTC Uniformity = 100 x [Total Parts Processed through Uniformity – Non-conforming to uniformity specs] / Total Parts processed through Uniformity.

Data is to be reported by shift. A weekly summary of the data shall be prepared and forwarded to the Chrysler/Fiat Release Engineer and SQA. Upon request the assembler must be capable of indicating the Primary/Audit balancer combination and Uniformity machine used during the shift.

6.4 Specification Conformance Verification

All assemblies must be verified to be within specifications using accepted statistical techniques. On site measurement of characteristics for 100% of assemblies can be waived when using parts with characteristics controlled through accepted statistical techniques. Known out of specification modules must be used to test key quality processes. Module non-conformance parameters and levels are to be defined by Fiat / Chrysler Engineering and SQA.

6.5 Process Potential Study

At initial production start-up of the process, the supplier, in conjunction with Chrysler/Fiat Supplier Quality and Engineering specialists shall perform a Process Potential Study on important process characteristics. Refer to Fiat / Chrysler’s Process Planning and Audit Manual.

6.6 Process Capability Study

Following successful completion of the Process Potential Study, a Process Capability Study shall be performed as required by Chrysler/Fiat's SQA manual on process characteristics designated by Chrysler/Fiat's SQA and Engineering. Refer to Chrysler/Fiat's Process Planning and Audit Manual.

6.7 Equipment Isoplotting

Wheel / tire assembler is required to run quarterly isoplot studies and provide reports (and data if requested) to Chrysler/Fiat Engineering and SQA. Isoplot studies must be run for each primary and audit balancer, and for each uniformity machine. Unless notified, equipment must meet a minimum discrimination ratio of 7.0. Contact Release Engineer for standardized isoplot file.

6.8 New Equipment Verification

New equipment balancer verification requires a gauge repeatability AND accuracy test be completed per Chrysler/Fiat Process Planning and Audit manual in accordance to the Measurement Systems Analysis Reference manual. Do not use the master for the REPEATABILITY test. Use the balance master for the ACCURACY test. In addition, balancers must pass a Dyn-10 and Dyn-20 test.

6.9 Launch Plan

The Wheel / tire assembler is required to execute a launch plan to prove out process and equipment capability prior to the Volume Production build of a new vehicle program and also prior to the launch of a new component that is released not during a vehicle launch. This plan is in addition to a DVP+R, and will be developed by the Chrysler/Fiat Release Engineer with input from Chrysler/Fiat SQA and the wheel/tire assembler and will be supplied to the wheel/tire assembler prior to the VP build. In the case of releasing new components not during a new vehicle launch, a modified plan will need to be created and executed. This plan includes, but is not limited to, running a statistically significant sample (quantity to be determined by Chrysler/Fiat Engineering) of the various wheel/tire assembly combinations in order to achieve conformance to balance and uniformity (if required) specifications and prove out assembly process capability. A statistical summary and all data must be provided to Chrysler/Fiat Engineering prior to launch and must clearly note any non-conformance or processing issues.

NOTE: All components must be integrated into the assembly line prior to submitting summary to Chrysler/Fiat Engineering.

7 AUDITS AND INSPECTIONS

7.1 Assembly Mounting Audit

The wheel and tire assembler will conduct wheel and tire module assembly audits to ensure that their assembly process does not damage any of the components in the wheel and tire assembly.

7.1.1 Audit Frequency

The tire mounting inspection will be performed at two frequencies: start-up frequency and standard frequency. Start-up frequency requires each shift to disassemble and inspect one assembly of each wheel and tire combination assembled during that shift when there is any of the following conditions:

- New tire/wheel combination or valve stem.
- Equipment repair or adjustment to any of the following: Valve stemmer, Tire-to-wheel mounter,
- Tire bead soaper or Wheel soaper.
- Downtime of five or more consecutive days.
- Tire or wheel supplier plant/process change.
- Wheel/tire assembler process change.

Different colors of the same wheel style (chrome wheels excluded) are not counted as additional combinations. Start-up frequency must continue issue-free for a minimum of two production shifts. The inspection can then be decreased to the standard frequency. Standard frequency requires the disassembly and inspection of a minimum of two different assemblies per shift. Wheel / tire assembly combinations must be rotated to encompass all possible combinations in production. Example of a wheel / tire assembly rotation is as follows:

Seven different tire/wheel combinations available (A, B, C, D, E, F, and G):
Day one, shift one: A, B
Day one, shift two: C, D
Day two, shift one: E, F
Day two, shift two: G, A

Every wheel / tire combination must be inspected at a minimum of once every two weeks.

7.1.2 Inspection Aids

The assembly inspection area shall include a permanent display of visual examples or photographs. See Figure 5. This display at a minimum shall include:

- Tires with cut beads,
- Tires with kinked or stretched beads,
- Tires with side-wall damage,
- Cosmetic damage to A-surface of wheel,
- Tire with excessive fluid accumulation

The inspection area must provide adequate task lighting to perform the following inspection process.

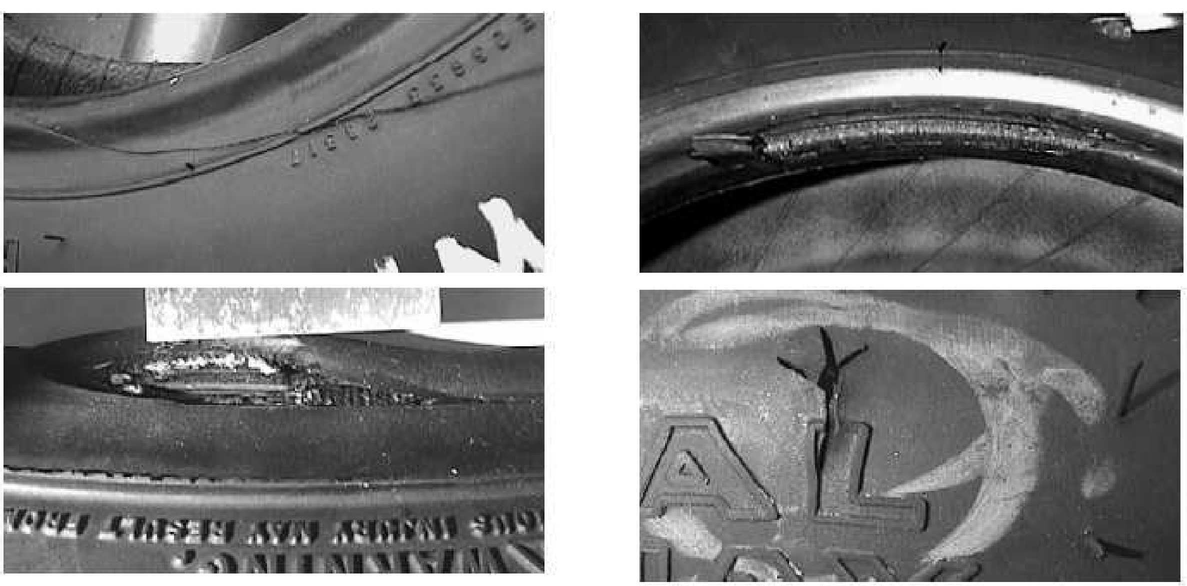

Figure 5 - Examples Of Tire Damage

7.1.3 Visual Inspection – Full Assembly

Remove the wheel / tire assembly from the end of the production line and transfer it to the inspection area. Visually inspect the wheel / tire assembly (exterior, prior to dismount) for any functional or cosmetic issues such as:

A. Tire bulges or excessive undulations <S>
B. Tire side wall scoring or cut <S>
C. Wheel and Center Cap (if applicable) scratches or scoring (See AS-10136)
D. Improper mounting (any irregularities or gaps in the bead mounting area of wheel and tire)
E. Balance weight not fully seated
F. Incorrect type of balance weight
G. Incorrect balance weight location
H. Improper match mount or missing high point mark
I. Center cap not fully seated
J. Improper placement of protective wheel film (if applicable)
K. Improper installation of valve stem cap

If the assembly includes either A, B, or C conditions, then the appropriate Supplier Quality Engineer must be contacted. For conditions D thru K or other visual issues, contact the appropriate plant supervisor immediately for corrective action. Upon concurrence, initiate the proper defective material procedure.

7.1.4 Valve Stem / TPM Inspection – Air Leakage <S>

Inspect the valve stem for proper seating and insure the stem has not been cut. For an assembly with a rubber valve, deflect the valve stem 30 degrees and articulate 360 degrees to check for air leakage with the lubricant specified in MS-9990 or suitable alternative such as P3 Tire Lube 40. See Figure 6. For an assembly with a metal valve stem or tire pressure sensor, check for air leakage with the lubricant specified in MS-9990 or suitable alternative. Do not attempt to deflect the valve stem. Any bubble appearing within 15 seconds indicates leakage and constitutes an unacceptable condition. If leakage is evident, contact the appropriate Supplier Quality Engineer immediately for corrective action. Upon concurrence, initiate the proper assembly defective material procedure. Refer to PS-11343 for further information on TPM valve stem inspection.

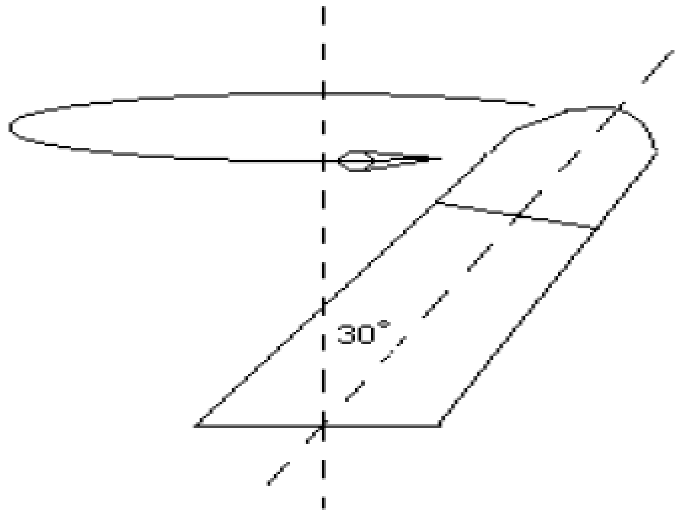

Figure 6 - Valve Stem Inspection

7.1.5 Verification of Air Pressure

Verify actual tire pressure per section 4.8. If the tire pressure is not within the tolerance of the specified inflation pressure initiate immediate corrective action. Tire pressure must be within specification before shipping the assembly into the vehicle plant.

Immerse assembly in water tank (if required) so that it is fully submerged for 60 seconds and check for air leakage evidenced by continuous flow of bubbles. No air leakage is accepted.

7.1.6 Tire Inspection

Disassemble the tire from the wheel using the disassembly procedure outlined in section 5.2.3. Thoroughly inspect the entire inner and outer circumference of the inboard and outboard tire bead. This inspection should include visual as well as tactile, i.e. flex the tire bead as shown in Figure 7 to expose any cut or tear. A cut or tear when flexed will expose rubber beneath the surface of the tire. Any cut or tear visible upon inspection without magnification is unacceptable. Inspect the outer and inner tire bead for deformation such as kinking due to excessive stretching. Any bead deformation is unacceptable. Thoroughly inspect the wheel rim for damage. Any visible wheel rim damage upon inspection without magnification is unacceptable. <S>

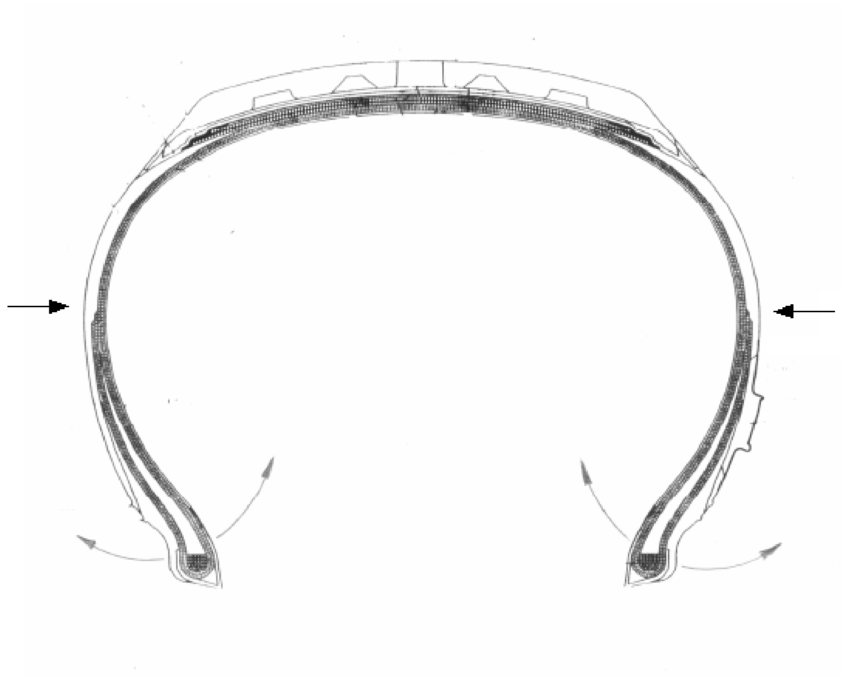

Figure 7 - Tire Inspection

If tire or wheel damage is found, contact the appropriate Supplier Quality Engineer immediately for corrective action. The above inspection procedure must be increased to hourly on this wheel style and tire combination until all sequential inspections are issue-free for all production shifts (all shifts proven to be clean). Different colors (chrome, and polished are considered unique colors) of the same wheel style are not counted as additional combinations.

In addition, the affected vehicles (VINS between the prior clean inspection point for that combination and the last inspection in which an issue was found) should be identified and held for disposition by Chrysler/Fiat Engineering. All disassembled tires and wheels in which an issue was found must also be held for further investigation by Chrysler/Fiat Engineering. The standard frequency of inspection for all other assemblies must remain at a minimum of two inspections per shift.

7.1.7 Fluid Accumulation, Foreign Objects

Inspect the inside of the tire for fluid accumulation. To determine the amount of fluid in a tire, first take a clean dry sponge and weigh it on a scale. Take the sponge and thoroughly wipe around the entire cavity of the tire. Weigh the wet sponge and then determine the weight of the fluid. The maximum allowable residual fluid is 0.25oz (7.1 grams) by weight. If no visible fluid exists then the fluid collection and weighing steps can be bypassed, but regular inspections must still occur. If fluid accumulation above 0.25 oz (7.1 grams) by weight is measured, contact the appropriate Supplier Quality Engineer immediately for corrective action. See Figure 8.

Figure 8 - Fluid Accumulation Not To Exceed 0.25oz By Weight As Shown Above
Reference Tire Shown Above Is A P225/55R17.

Inspect the inside of the tire for other foreign material accumulation. Contact the appropriate assembly plant vehicle supervisor if excessive foreign material is found.

7.1.8 Wheel to Valve Stem / TPM Inspection<S>

Inspect for proper seating of the valve stem. For TPM inspection, refer to PS-11343. For rubber valve stems, remove the valve stem and inspect for scratches, scoring or other damage in the valve hole area. If improper valve seating is found, contact the appropriate vehicle assembly plant supervisor immediately for corrective action. Upon concurrence, initiate the proper assembly plant issue material procedure. <S>

7.1.9 Reinsertion of Undamaged Audit Components

Issue-free wheels and tires can be reintroduced to the start of the wheel and tire assembly process. It is permissible for reprocessed flangeless wheels to have adhesive weight residue on the application area of the wheel, however it is the wheel/tire assembler's responsibility to remove as much of the residue as reasonably possible before applying additional weights. No visible residue is allowed in the windows between the wheel spokes.

7.2 Tire Slippage Audit

The balance weight on the wheel and the mark on the tire used to locate the wheel weight must be within 25 mm of their geometric centers, as shown below in Figure 9, when measured along the wheel's rim flange, to verify that no tire slippage has occurred. This procedure is to be used for tire slippage unless another audit requirement is specified by Chrysler/Fiat Engineering. The Wheel / Tire Assembler must support resolution of any tire slippage issues identified at the vehicle assembly plant.

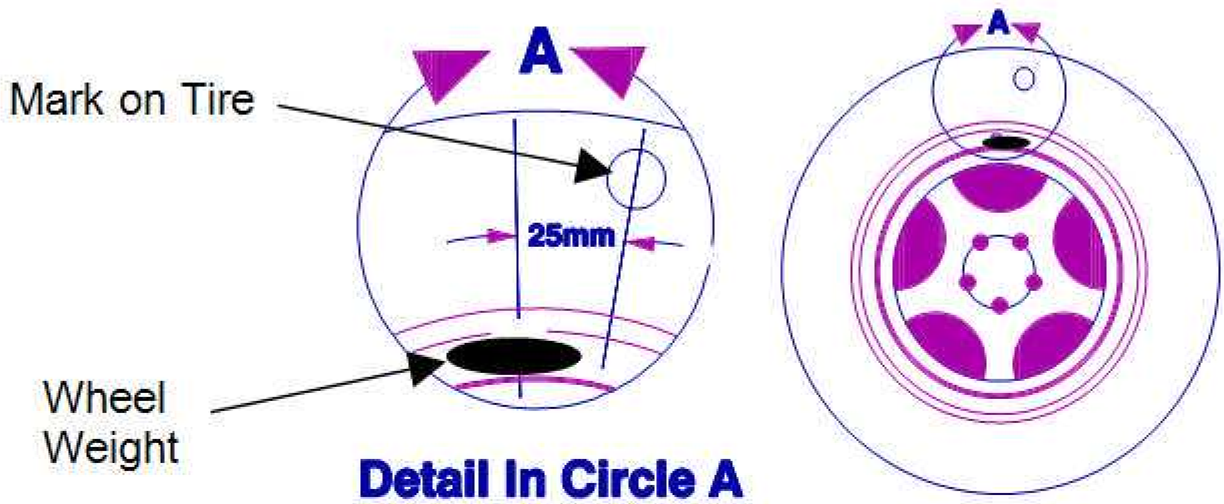

Figure 9 - Tire Slippage Audit

NOTE: For assemblies containing Euro flange wheels, for which the balance weight locating marks are applied to the inboard side of the tire, a chalk mark must be manually applied on the tire's outboard sidewall above the valve hole. The chalk mark and valve hole will then be used as substitutes to determine tire slippage. With approval from Chrysler/Fiat Engineering, the wheel/assembler may use this method also for marking assemblies containing MC-flange wheels.

8 DISPOSITION OF REJECTED COMPONENTS

Within 5 business days of deeming a component to be non-conforming, the wheel/tire assembler must contact the appropriate plant SQE for non-conforming product or plant production control manager for adjustment of inventory. Assembler is not allowed to scrap out any consigned component without permission from plant production control.

9 TEST EQUIPMENT

No special test equipment required

10 SAFETY PRECAUTIONS

Use normal safety precautions when working around heavy parts and high loading conditions. Follow all appropriate plant safety rules and practices.

11 APPROVED SOURCE LIST

Not Applicable.